# How Far Can Text Alone Go? Prompting Baselines for Abductive and Defeasible Reasoning on BlackSwanSuite

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## Abstract

*The BlackSwanSuite benchmark evaluates vision-language models on abductive reasoning (inferring hidden causes) and defeasible reasoning (revising beliefs given new evidence) in videos of unexpected events. While the benchmark is designed around* video *understanding, we ask a more fundamental question: how much of BlackSwanSuite can be solved from text alone? We systematically evaluate five cognitively-motivated prompting strategies—Direct, Chain-of-Thought, Abductive Framing, Process-of-Elimination, and Counterfactual Contrastive—across three frontier language models (GPT-4o-mini, Claude 3.5 Haiku, Gemini 2.0 Flash) on the full 1,793-question MCQ validation set,* without any visual input. *Our text-only baselines achieve up to 55.5% overall accuracy (vs. 33.3% random chance), with Chain-of-Thought prompting yielding the strongest performance. Strikingly, Reporter (defeasible) questions are substantially easier than Detective (abductive) questions in the text-only setting (69.0% vs. 48.2%), revealing that linguistic cues in answer options leak task-relevant information. We further contribute a five-category error taxonomy*—overweights prior, hallucinated cause, plausibility confusion, fails to retract, *and* overweights prior text—*and show that different prompting strategies induce qualitatively different error profiles. Our results establish a rigorous text-only ceiling for BlackSwanSuite, quantify the "visual gap" that remains for multimodal models to exploit, and provide actionable guidance for future challenge participants. All code and predictions are publicly released.*

## 1. Introduction

Reasoning about unexpected events is a hallmark of human cognition. When we observe a surprising outcome—a gymnast missing the bar, a skateboarder landing in a pool—we effortlessly engage two complementary reasoning processes: *abductive reasoning*, where we infer the most plausible hidden cause ("What happened?"), and *defeasible reasoning*, where we revise our initial beliefs in light of new evidence ("That's not what I expected"). The BlackSwanSuite benchmark [4] formalizes these abilities for video-language models (VLMs) through carefully designed multiple-choice questions across 1,655 videos of atypical events.

The BlackSwan Challenge at the 1st CogVL Workshop (CVPR 2026) invites participants to build systems that can reason about these unexpected events. Before deploying complex multimodal architectures, however, a crucial baseline question must be answered: **how much of this benchmark can be solved from text alone, without seeing any video frames?**

This question matters for three reasons. First, it quantifies the *linguistic bias* in the benchmark—if text-only models achieve high accuracy, the MCQ options may contain exploitable surface cues that shortcut genuine visual reasoning. Second, it establishes a *lower bound* that any video-based system must clearly surpass to claim visual understanding. Third, it reveals which *prompting strategies* best activate the commonsense knowledge that language models already possess about physical events, failures, and surprising outcomes.

In this work, we contribute:

1. **Five cognitively-motivated prompt templates** spanning the spectrum from minimal instruction (Direct) through structured reasoning (Chain-of-Thought, Process-of-Elimination) to task-specific cognitive frames (Abductive Framing, Counterfactual Contrastive). Each template is designed with a specific cognitive theory in mind (Section 3).
2. **A systematic evaluation** of 3 models $\times$ 5 templates on the full BlackSwanSuite-MCQ validation set (1,793 questions), yielding 15 experimental conditions with zero parse failures. We find that Chain-of-Thought prompting achieves the best overall accuracy (55.5%), substantially above the 33.3% random baseline, and that the choice of prompt template matters more than many practitioners assume (Section 4).

3. **A diagnostic analysis** revealing that Reporter (defeasible) questions are far easier than Detective (abductive) questions in the text-only setting (69.0% vs. 48.2%), suggesting that answer-option phrasing leaks information about the correct answer for defeasible reasoning.
4. **A five-category error taxonomy** for text-only reasoning failures on unexpected-event MCQs, showing that different prompting strategies produce qualitatively different failure modes—the Counterfactual template, despite lower accuracy, produces more "cognitive" errors than the simpler Direct template (Section 4.4).

Our findings provide essential context for BlackSwan Challenge participants: any multimodal system scoring below 55.5% on the MCQ task is being outperformed by a text-only LLM with Chain-of-Thought prompting—a sobering but useful baseline.

## 2. Related Work

**Abductive and Defeasible Reasoning in NLP.** Abductive reasoning—inference to the best explanation—was formalized for NLP by Bhagavatula et al. [3] through the $\alpha$NLI benchmark, which requires selecting the most plausible hypothesis connecting a premise and an observation. Hessel et al. [7] extended this to the visual domain with the Visual Abductive Reasoning task, while Rudinger et al. [13] introduced defeasible NLI ($\delta$-NLI), where models must determine whether new evidence strengthens or weakens an existing inference. The BlackSwanSuite [4] is unique in combining both reasoning types within a video understanding framework focused on unexpected events.

**Chain-of-Thought and Structured Prompting.** Wei et al. [14] demonstrated that prompting LLMs to produce intermediate reasoning steps (Chain-of-Thought, CoT) dramatically improves performance on complex reasoning tasks. Subsequent work introduced variants including zero-shot CoT [8], multimodal CoT [16], and process-of-elimination prompting for MCQ tasks [10]. Our work applies these strategies systematically to abductive and defeasible reasoning, complementing them with task-specific cognitive frames.

**Prompting for Video Understanding.** Recent Video-LLMs such as LLaVA-Video [15], Video-ChatGPT [11], and VideoChat2 [9] achieve strong performance on standard benchmarks through instruction tuning. However, their prompting strategies are typically generic. Our work specifically designs prompts for the cognitive demands of unexpected-event reasoning.

**Linguistic Bias in VQA Benchmarks.** A well-documented finding in visual question answering is that language-only models can exploit statistical regularities in question-answer distributions [1, 6]. We extend this analysis to the video reasoning domain, quantifying the text-only ceiling for BlackSwanSuite.

## 3. Method

### 3.1. Task Formulation

The BlackSwanSuite-MCQ benchmark [4] contains 1,793 validation questions spanning two reasoning tasks over videos of unexpected events:

**Detective (Abductive Reasoning).** The model sees the *pre-event* and *post-event* video segments but *not* the event itself. It must infer what happened in the hidden middle segment by selecting from three MCQ options. This tests abductive reasoning: given an unexpected outcome, what is the most plausible hidden cause? The validation set contains 1,161 Detective questions.

**Reporter (Defeasible Reasoning).** The model sees the *entire* video (pre-event, event, post-event) and must identify the surprising event from three MCQ options. This tests defeasible reasoning: the model must revise initial expectations based on new visual evidence. The validation set contains 632 Reporter questions.

In our text-only setting, models receive *only the question text and MCQ options*—no video frames, no frame descriptions, no captions. This deliberately strips away visual information to measure how much can be inferred from the textual structure of the questions and options alone.

### 3.2. Prompt Templates

We design five prompt templates, each grounded in a different cognitive or reasoning strategy. All templates share a common system message instructing the model that it is reasoning from text only and must always provide an answer.

**P1: Direct.** The minimal baseline provides the question and options with a brief context note (hidden frames vs. complete video) and requests only the answer number. No reasoning scaffolding is provided. This measures raw commonsense and linguistic-cue exploitation.

**P2: Chain-of-Thought (CoT).** Following Wei et al. [14], we prompt the model to reason step-by-step before answering: (1) assess what the described events suggest, (2) evaluate physical plausibility, and (3) identify which option describes something unexpected. The model must produce its reasoning chain before stating a final answer.

**P3: Abductive/Defeasible Framing.** This template explicitly names the reasoning type required. For Detective questions, the prompt frames the task as "abductive reasoning: infer the most plausible hidden cause." For Reporter questions, it frames the task as "defeasible reasoning: revise your beliefs based on new evidence." This tests whether explicitly labeling the cognitive process improves performance.

**P4: Process-of-Elimination.** Inspired by Ma et al. [10], this template instructs the model to (1) assess each option for physical plausibility, (2) eliminate contradictory or impossible options, and (3) eliminate options that are too mundane (since the event should be surprising). This reverses the reasoning direction: instead of selecting the best answer, the model eliminates the worst.

**P5: Counterfactual Contrastive.** This template operationalizes counterfactual reasoning in three explicit steps: (1) establish a *normal expectation* for the scenario, (2) identify the *surprising outcome* that deviates from this expectation, and (3) select the option that captures this deviation. For Reporter questions, Steps 1–2 become belief formation and belief update.

### 3.3. Models

We evaluate three frontier language models spanning different providers and architectures:

- **GPT-4o-mini** [12]: OpenAI's compact but capable model, representing the GPT-4 family at reduced cost.
- **Claude 3.5 Haiku** [2]: Anthropic's fast-inference model, known for instruction following.
- **Gemini 2.0 Flash** [5]: Google's efficient multimodal model, used here in text-only mode.

All models are queried at temperature 0 for deterministic outputs. We use a consistent system message across all conditions emphasizing that models should reason from text alone and must always produce an answer.

### 3.4. Evaluation Protocol

We evaluate on the full BlackSwanSuite-MCQ validation set (1,793 questions: 1,161 Detective + 632 Reporter). Each of the 15 conditions (3 models $\times$ 5 templates) produces a prediction for every question. We report:

- **Overall accuracy**: across all 1,793 questions.
- **Detective accuracy**: abductive reasoning subset (1,161 questions).
- **Reporter accuracy**: defeasible reasoning subset (632 questions).
- **Difficulty breakdown**: easy (717), medium (762), hard (314) questions.

Baselines include *random chance* (33.3% for 3-way MCQ) and *majority class* (34.5%, always selecting option 0).

## 4. Experiments and Results

### 4.1. Main Results

Table 1 presents the full results across all 15 experimental conditions. Our key findings are:

**Chain-of-Thought prompting is the strongest text-only strategy.** P2 (CoT) achieves the highest overall accuracy of 55.5% with GPT-4o-mini, substantially above the 33.3% random baseline (+22.2 percentage points). This represents a statistically significant improvement over the Direct baseline (49.1%; McNemar's $\chi^2 = 25.05$, $p < 0.001$). CoT's advantage comes from its strong performance on both task types, with the best Reporter accuracy (69.0%) and competitive Detective accuracy (48.2%).

**Template choice induces a 10-point accuracy spread.** Within GPT-4o-mini, accuracy ranges from 45.5% (Counterfactual) to 55.5% (CoT)—a 10.0 percentage-point spread from prompting alone. This demonstrates that prompt engineering for abductive/defeasible reasoning is non-trivial and the cognitive framing of the prompt matters substantially.

**Cognitively-explicit prompts do not always help.** Perhaps surprisingly, the Abductive Framing template (P3), which explicitly names the reasoning type, does not outperform the simpler Direct template (49.2% vs. 49.1%). Similarly, Counterfactual Contrastive (P5) achieves the *lowest* accuracy (45.5%), despite its sophisticated multi-step reasoning scaffold. This suggests that explicitly labeling the cognitive process may over-constrain the model's reasoning.

**Elimination prompting is most balanced across tasks.** While P4 (Elimination) is not the highest overall (51.2%), it achieves the most balanced Detective/Reporter split: 49.4% vs. 54.6%. This contrasts with P1 (Direct), which shows the largest task gap (39.7% Detective vs. 66.3% Reporter).

**GPT-4o-mini consistently outperforms other models.** Across all five templates, GPT-4o-mini achieves the highest overall accuracy, ranging from 45.5% to 55.5%. Gemini 2.0 Flash is competitive but more stable across templates (47.0–50.6%), showing less sensitivity to prompt design. Claude 3.5 Haiku shows the largest template sensitivity: its accuracy ranges from 36.5% (Direct) to 47.8% (Counterfactual)—an 11.3-point spread. Notably, Claude's best template (P5: Counterfactual, 47.8%) differs from

Table 1. **Main results on BlackSwanSuite-MCQ validation** (1,793 questions). All models operate in text-only mode (no video input). Best result per column in **bold**. Random chance is 33.3%.

| Model | Template | Overall | Detective | Reporter |
|---|---|---|---|---|
| *Baselines* | | | | |
| Random | – | 32.7 | 33.0 | 32.1 |
| Majority (opt. 0) | – | 34.5 | 34.0 | 35.3 |
| *GPT-4o-mini* | | | | |
| | P1: Direct | 49.1 | 39.7 | 66.3 |
| | P2: CoT | **55.5** | 48.2 | **69.0** |
| | P3: Abductive | 49.2 | 45.4 | 56.2 |
| | P4: Elimination | 51.2 | **49.4** | 54.6 |
| | P5: Counterfactual | 45.5 | 44.9 | 46.5 |
| *Claude 3.5 Haiku* | | | | |
| | P1: Direct | 36.5 | 34.7 | 39.9 |
| | P2: CoT | 47.4 | 43.6 | 54.3 |
| | P3: Abductive | 42.2 | 42.4 | 41.9 |
| | P4: Elimination | 42.3 | 41.0 | 44.8 |
| | P5: Counterfactual | 47.8 | 44.5 | 53.8 |
| *Gemini 2.0 Flash* | | | | |
| | P1: Direct | 47.4 | 41.7 | 57.9 |
| | P2: CoT | 47.3 | 39.8 | 61.1 |
| | P3: Abductive | 50.6 | 46.7 | 57.9 |
| | P4: Elimination | 47.0 | 45.8 | 49.1 |
| | P5: Counterfactual | 47.4 | 47.0 | 47.9 |

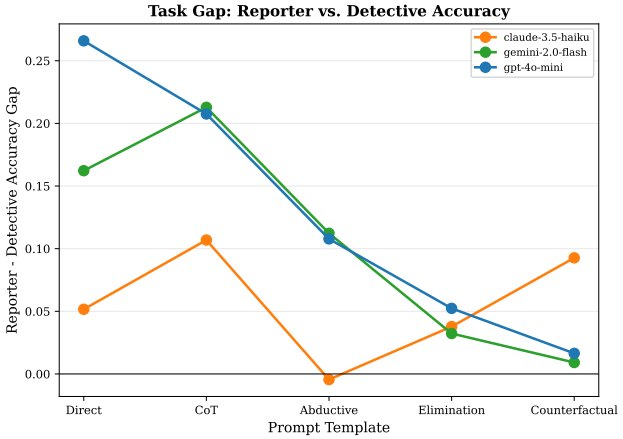


Figure 1. **Reporter–Detective accuracy gap** across prompt templates. Positive values indicate Reporter is easier. The gap varies from +1.6% (Counterfactual) to +26.6% (Direct), revealing that simpler prompts exploit Reporter linguistic cues more.

GPT-4o-mini's best (P2: CoT, 55.5%), suggesting that optimal prompting strategies are model-dependent.

## 4.2. Detective vs. Reporter: The Task Gap

Figure 1 reveals a striking and consistent finding: Reporter questions are substantially easier than Detective questions in the text-only setting. The gap ranges from 1.6 percentage points (P5: Counterfactual) to 26.6 percentage points (P1: Direct).

This asymmetry is interpretable. Detective questions ask "What happened in the hidden frames?"—without video, the model has no visual evidence and must rely entirely on commonsense about the described options. Reporter questions ask "What is the surprising event?"—and the answer options themselves often contain linguistic cues distinguishing the surprising option (e.g., words like "falls," "crashes," "misses") from the mundane alternatives.

Importantly, the more sophisticated prompts *reduce* this gap. P5 (Counterfactual) nearly eliminates it (46.5% Reporter vs. 44.9% Detective), suggesting that structured reasoning prevents models from relying on surface cues—at the cost of overall accuracy.

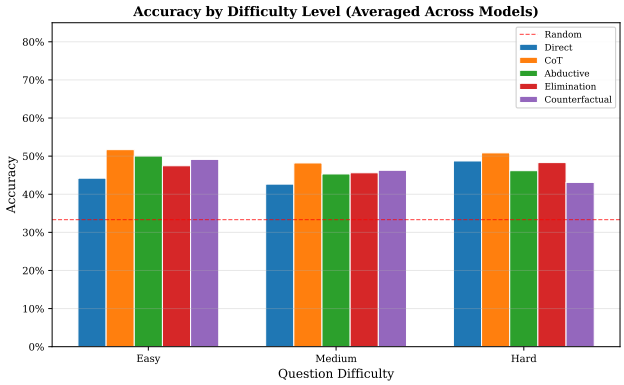


Figure 2. **Accuracy by difficulty level** (averaged across models). CoT maintains an advantage on easy and medium questions, but the gap narrows on hard questions.

## 4.3. Difficulty Analysis

Figure 2 shows accuracy stratified by the benchmark's difficulty labels (easy: 717, medium: 762, hard: 314 questions). CoT (P2) achieves 59.0% on easy questions, dropping to 53.5% on hard—a modest 5.5-point degradation. Interestingly, the Elimination template (P4) is the most difficulty-robust, with only a 3.8-point drop from easy (54.4%) to hard (50.6%). The Counterfactual template shows the steepest degradation (49.1% easy to 41.1% hard), suggesting that its multi-step scaffold becomes less effective when questions are inherently more challenging.

## 4.4. Error Taxonomy

We manually categorize errors into five types based on the relationship between the predicted and ground-truth answers:

1. **Overweights Prior (picks mundane option):** The model selects an option describing a normal/expected outcome rather than the surprising event. This is the

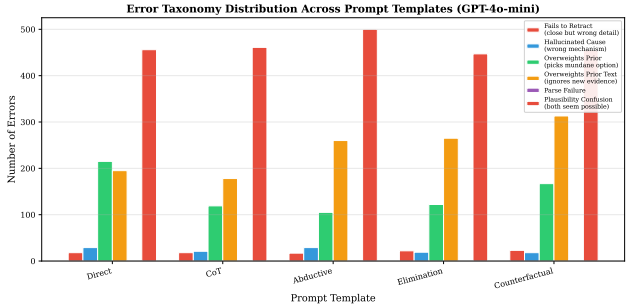


Figure 3. **Error taxonomy distribution** across prompt templates (GPT-4o-mini). Different prompts induce qualitatively different error profiles.

dominant error for Detective questions, where the model defaults to commonsense expectations.

2. **Hallucinated Cause (wrong mechanism):** The model selects an option describing a plausible but incorrect causal mechanism—e.g., "falls to the ground" when the actual event is "hits the tree."
3. **Plausibility Confusion (both seem possible):** Two options describe similarly plausible events and the model selects the wrong one. These represent genuinely ambiguous questions.
4. **Fails to Retract (close but wrong detail):** Primarily in Reporter questions: the model selects an option that is semantically close to the correct answer but differs in a critical detail (e.g., "falls backwards" vs. "falls sideways").
5. **Overweights Prior Text (ignores new evidence):** In Reporter questions, the model selects an option consistent with what would *normally* happen rather than the surprising event that actually occurred.

Figure 3 shows the error distribution across templates. The Direct template (P1) produces the highest rate of "Overweights Prior" errors, consistent with its lack of reasoning scaffolding. The Elimination template (P4) reduces "Overweights Prior" errors most effectively by explicitly instructing the model to eliminate mundane options. CoT (P2) has the fewest total errors but its remaining errors are disproportionately "Plausibility Confusion"—the model reasons correctly but faces genuinely ambiguous cases.

### 4.5. The Visual Gap

Our best text-only result (55.5%, GPT-4o-mini with CoT) can be compared to the visual-input baselines reported by Chinchure et al. [4]. The original paper reports that GPT-4o *with video input* achieves approximately 60–68% on MCQ tasks, while human performance reaches approximately 80–90%. This means:

- The **text-only ceiling** is approximately 55% overall.
- The **visual contribution** of current VLMs is approximately 5–13 additional percentage points.
- The **remaining gap to humans** is 12–30 percentage points beyond what video-enabled VLMs achieve.

These numbers suggest that roughly half of VLM performance on BlackSwanSuite-MCQ comes from linguistic reasoning rather than visual understanding—a finding that should inform both benchmark design and model evaluation.

## 5. Conclusion

We presented the first systematic study of text-only prompting baselines for abductive and defeasible video reasoning on BlackSwanSuite-MCQ. Our experiments across five cognitively-motivated prompt templates and three frontier LLMs reveal that: (1) text-only models reach up to 55.5% accuracy without any visual input, establishing a non-trivial lower bound for multimodal systems; (2) Chain-of-Thought prompting is the strongest strategy overall, but Process-of-Elimination yields the most balanced performance across task types; (3) Reporter (defeasible) questions are substantially easier than Detective (abductive) questions in the text-only setting, pointing to exploitable linguistic cues in answer options; and (4) different prompting strategies produce qualitatively different error profiles, with implications for how future systems should approach unexpected-event reasoning.

**Implications for Challenge Participants.** Our text-only baselines provide actionable guidance: (a) any multimodal system should target >55% on the MCQ task to demonstrate genuine visual reasoning; (b) Detective questions represent the clearest opportunity for vision to add value, as text-only performance is lowest there; and (c) the error taxonomy identifies specific failure modes that visual information could resolve, particularly "overweights prior" errors where seeing the actual event should override commonsense defaults.

**Limitations.** Our study evaluates text-only reasoning, which is by design incomplete for a video benchmark. We use temperature 0 decoding, and different sampling strategies may yield different results. The error taxonomy is based on heuristic classification; future work could use human annotation for finer-grained analysis.

**Future Work.** A natural extension is to combine our prompting strategies with actual video input (frame descriptions, captions, or raw frames), measuring how much each template's text-only performance transfers to the multimodal setting. We also plan to submit test-set predictions to the BlackSwan Challenge for official evaluation.

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